

Experiment-8

A)Aim:Write a program for Conditional Rendering

Description:

Conditional rendering is a technique in React used to display different user interface elements based on certain conditions.

It allows the application to dynamically change what is shown on the screen depending on the state, props, or user actions.

In simple words, conditional rendering means showing or hiding components based on a condition, like using if-else statements in JavaScript.

Program:

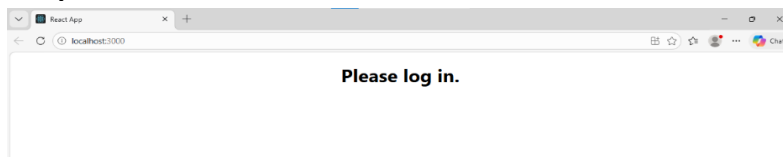
Using if Statement

```
function Greeting({ isLoggedIn }) {
  if (isLoggedIn) {
    return <h1>Welcome back!</h1>;
  }
  return <h1>Please log in.</h1>;
}

function App() {
  return (
    <div className="App">
      <Greeting isLoggedIn={false} />
    </div>
  );
}

export default App;
```

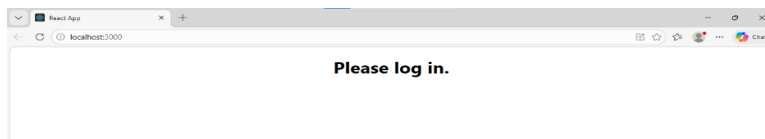
Output:



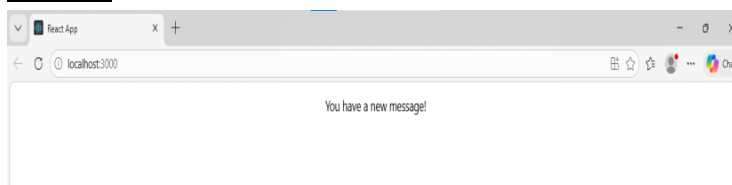
Using Ternary Operator (? :)

```
function Greeting({ isLoggedIn }) {
  return (
    <div>
      {isLoggedIn ? <h1>Welcome back!</h1> : <h1>Please log in.</h1>}
    </div>
  );
}
```

```
function App() {
  return (
    <div className="App">
      <Greeting isLoggedIn={false} />
    </div>
  );
}
export default App;
```

Output:**Using Logical AND (&&)**

```
function Notification({ hasMessage }) {
  return (
    <div>
      {hasMessage && <p>You have a new message!</p>}
    </div>
  );
}
function App() {
  return (
    <div className="App">
      <Notification hasMessage={true} />
    </div>
  );
}
export default App;
```

Output:**Using Element Variables**

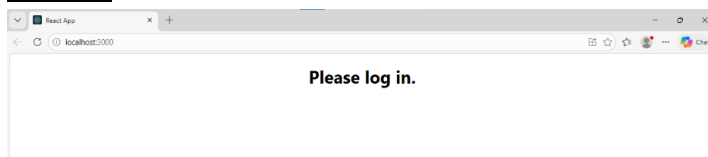
```
function Greeting({ isLoggedIn }) {
  let message;
  if (isLoggedIn) {
    message = <h1>Welcome back!</h1>;
  }
}
```

```

    } else {
      message = <h1>Please log in.</h1>;
    }
    return <div>{message}</div>;
  }
function App() {
  return (
    <div className="App">
      <Greeting isLoggedIn={true} />
    </div>
  );
}

```

Output:



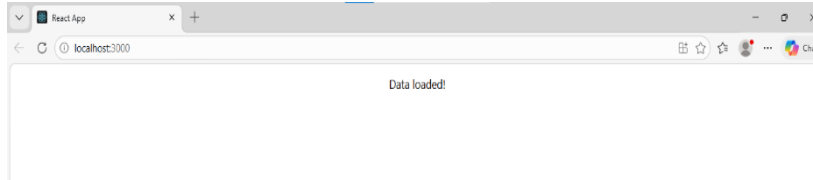
Using Switch Statement

```

function Status({ status }) {
  switch (status) {
    case "loading":
      return <p>Loading...</p>;
    case "success":
      return <p>Data loaded!</p>;
    case "error":
      return <p>Error occurred.</p>;
    default:
      return null;
  }
}
function App() {
  return (
    <div className="App">
      <Status status={"success"} />
    </div>
  );
}
export default App;

```

Output:



Conditional Rendering with Components

Program:

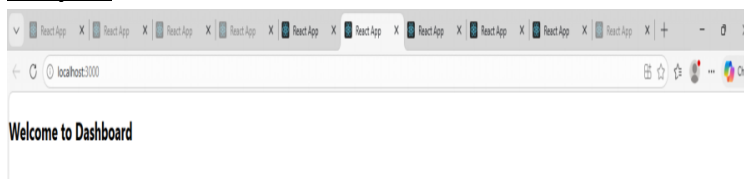
```
import './App.css';

function Dashboard() {
  return <h2>Welcome to the Dashboard!</h2>;
}

function Login() {
  return <h2>Please log in.</h2>;
}

function App() {
  const isLoggedIn = false;
  return (
    <div>
      {isLoggedIn && <Dashboard />}
      {!isLoggedIn && <Login />}
    </div>
  );
}
```

Output:



B)Aim:Write a program for rendering lists

Description:

In modern web applications, we often need to display many items such as a list of users, products, messages, or files.

This process is called Rendering Lists.

React uses the JavaScript `map()` function to go through each element of the array and create a UI element for every item.

Program:

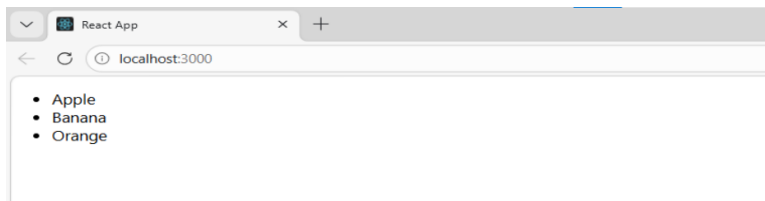
```
function FruitList() {
  const fruits = ["Apple", "Banana", "Orange"];
  return (
```

```

    <ul>
      {fruits.map((fruit, index) => (
        <li key={index}>{fruit}</li>
      ))}
    </ul>
  );}
function App() {
  return (
    <div className="App">
      <FruitList />
    </div>
  );}

```

Output:

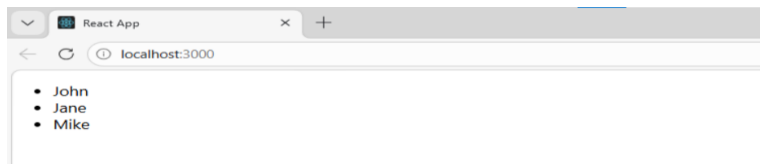


Rendering List of Objects

```

function UserList() {
  const users = [
    { id: 1, name: "Sai" },
    { id: 2, name: "Krish" },
    { id: 3, name: "Teja" }
  ];
  return (
    <ul>
      {users.map(user => (
        <li key={user.id}>{user.name}</li>
      ))}
    </ul>
  );}
function App() {
  return (
    <div className="App">
      <UserList />
    </div>
  );}
export default App;

```

Output:**Why key is Important?**

React uses key to:

- Identify which items changed
- Improve performance
- Prevent unnecessary re-renders

8c)Aim: Write a program for working with different form fields using react forms

Rendering Components in a List

```
function Product({ name, price }) {
  return (
    <div>
      <h3>{name}</h3>
      <p>${price}</p>
    </div>
  );
}

function ProductList() {
  const products = [
    { id: 1, name: "Laptop", price: 1000 },
    { id: 2, name: "Phone", price: 500 },
    { id: 3, name: "Watch", price: 1500 },
  ];
  return (
    <div>
      {products.map(product => (
        <Product
          key={product.id}
          name={product.name}
          price={product.price}
        />
      ))}
    </div>
  );
}

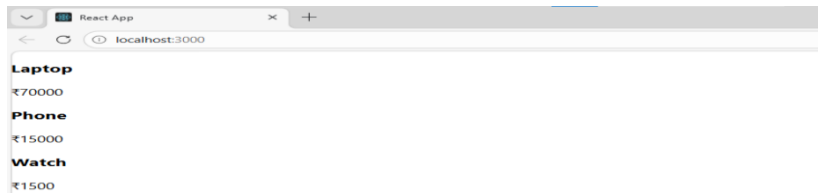
function App() {
  return (
    <div className="App">
```

```

    <ProductList/>
    <Product />
  </div>
);}
export default App;

```

Output:



Conditional Rendering Inside Lists

```

import React from "react";
function UserList() {
  const users = [
    { id: 1, name: "John", isActive: true },
    { id: 2, name: "Jane", isActive: false },
    { id: 3, name: "Mike", isActive: true }
  ];
  return (
    <ul>
      {users
        .filter(user => user.isActive)
        .map(user => (
          <li key={user.id}>{user.name}</li>
        ))}
    </ul>
  );}
function App() {
  return <UserList />;
}
export default App;

```

Output:

